

EB-JEPA — Reference Table (V1)

EB-JEPA EEG track

June 2026

Our EB-JEPA runs on TUAB (v3.0.1, patient-disjoint, 276 eval recordings). *per-rec* = per-recording (16-window mean-pool, the clinical metric); *per-win* = per-window (the literature convention). $\pm$ = 3-seed standard deviation; other rows are single-seed. Companion to `explanation_augmented_corruption.tex` (which details the augmentation) and `PROVENANCE_TABLE.md` (the literature baselines).

#	Encoder	Params	Energy (SSL objective)	Eval	per-rec BAcc / AUROC	per-win BAcc / AUROC
baseline	Conv1D	0.4M	VICReg — <i>base aug, no corruption</i>	frozen	0.796 / 0.888	0.756 / –
1	Conv1D	0.4M	VICReg + spectral 0.1	frozen	0.836 / 0.887	0.765 / –
2	Conv1D	0.4M	SIGReg	frozen	0.825 / 0.913	0.775 / 0.856
3	Conv1D	0.4M	VICReg	frozen	0.819 $\pm$.004 / 0.900 $\pm$.006	0.770 / 0.848
4	Conv1D	0.4M	VICReg	FT	0.812 $\pm$.004 / 0.908 $\pm$.006	–
5	Transformer (multi-corpus)	3.65M	VICReg	frozen	0.798 / 0.872	0.738 / –

Fixed for every corruption run (never swept): views v_1, v_2 made with Gaussian noise $\sigma = 0.1$, scale-jitter $\pm 20\%$, channel-drop $p = 0.2$, plus exact EB-corruption — 20% timepoints masked + 20% $\pm 6\sigma$ outliers ($\approx 40\%$ /view). The **baseline** is the no-corruption reference (legacy contiguous time-mask).